

UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

Data Structure (BCS301)

Roll No. _____

Subject: Data Structure

Time: 3 Hours

Max. Marks: 70

Code: BCS301

General Instructions:

1. Attempt all questions.
2. The total marks for each section are indicated.
3. Marks for each question are specified alongside.

SECTION A — SECTION A

Short Answer Type Questions

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	Define Abstract Data Type (ADT) with a suitable example.	2	CO1
2	State the best-case and worst-case time complexity of the Bubble Sort algorithm.	2	CO2
3	Define a Sparse Matrix and explain its triplet representation.	2	CO3
4	List the two fundamental conditions required for a recursive function to terminate.	2	CO4
5	Explain the primary advantage of a Circular Queue over a linear queue implementation.	2	CO5
6	Explain Big-Oh notation and its significance in algorithm analysis.	2	CO1
7	Compare Sequential Search and Binary Search based on their search space reduction strategy.	2	CO2

SECTION B — SECTION B*Attempt only 3 questions**7 marks each | Attempt all questions*

Q.No.	Question	Marks	CO
8	Explain the algorithm for Polynomial addition using a Singly Linked List representation.	7	CO3
9	Describe the process of evaluating a Postfix expression using a stack with a step-by-step procedure.	7	CO4
10	Discuss the implementation of a Priority Queue and explain the logic for its 'delete' operation.	7	CO5
11	Derive the address calculation formula for an element $A[i][j][k][l]$ in a 4-dimensional array stored in Row Major Order.	7	CO1
12	Explain the Quick Sort algorithm and provide a detailed analysis of its performance in the worst-case scenario.	7	CO2

SECTION C — SECTION C			
Choice Group Questions			
7 marks each Attempt all questions			
Q.No.	Question	Marks	CO
13	(a) Describe the insertion and deletion operations in a Doubly Linked List with the help of diagrams.	7	CO3
	OR		
	(b) Compare the efficiency and memory overhead of Singly Linked Lists, Doubly Linked Lists, and Circular Linked Lists.	7	CO3
14	(a) Analyze the recursive algorithm for the Tower of Hanoi problem and determine its total number of moves for 'n' disks.	7	CO4
	OR		
	(b) Explain the mechanism of removing recursion from a function using an explicit stack with a specific example.	7	CO4
15	(a) Discuss the implementation of a Double-Ended Queue (Deque) and explain the difference between input-restricted and output-restricted dequeues.	7	CO5
	OR		
	(b) Evaluate the performance trade-offs between array-based and linked-list-based implementations of a Queue data structure.	7	CO5
16	(a) Explain the concept of Time-Space trade-off in algorithms and provide an example where reducing space increases time complexity.	7	CO1
	OR		
	(b) Discuss the Column Major Order representation of a 3D array and derive the general formula for calculating the address of element $A[i][j][k]$.	7	CO1
17	(a) Explain the Merge Sort algorithm and prove that its time complexity is $O(n \log n)$ in all cases.	7	CO2
	OR		
	(b) Discuss various Collision Resolution techniques in Hashing, specifically comparing Chaining and Open Addressing methods.	7	CO2

Question Paper Analysis

Bloom's Taxonomy Distribution



- K1 – Remember: 6m (7%)
- K2 – Understand: 8m (10%)
- K3 – Apply: 14m (17%)
- K4 – Analyze: 42m (50%)
- K5 – Evaluate: 14m (17%)
- K6 – Create: 0m (0%)

Course Outcome Distribution

